

mind that the derivatives in Eqs. (7.5.5) and (7.5.6) are all evaluated at the equilibrium flight condition. Clearly, the axial force is not totally independent of sideslip. However, from the assumed symmetry, a positive sideslip should have the same effect on the axial force as a negative sideslip. Thus, the axial force must be a symmetric function of sideslip, and the change in axial force with respect to sideslip velocity must be zero, when the sideslip velocity is zero. By similar reasoning we can eliminate all of the following derivatives evaluated at the reference flight condition:

$$\begin{aligned}
 \frac{\partial F_{x_b}}{\partial v} &= \frac{\partial F_{y_b}}{\partial u} = \frac{\partial F_{y_b}}{\partial w} = \frac{\partial F_{z_b}}{\partial v} = 0 \\
 \frac{\partial M_{x_b}}{\partial u} &= \frac{\partial M_{x_b}}{\partial w} = \frac{\partial M_{y_b}}{\partial v} = \frac{\partial M_{z_b}}{\partial u} = \frac{\partial M_{z_b}}{\partial w} = 0 \\
 \frac{\partial F_{x_b}}{\partial p} &= \frac{\partial F_{x_b}}{\partial r} = \frac{\partial F_{y_b}}{\partial q} = \frac{\partial F_{z_b}}{\partial p} = \frac{\partial F_{z_b}}{\partial r} = 0 \\
 \frac{\partial M_{x_b}}{\partial q} &= \frac{\partial M_{y_b}}{\partial p} = \frac{\partial M_{y_b}}{\partial r} = \frac{\partial M_{z_b}}{\partial q} = 0 \\
 \frac{\partial F_{x_b}}{\partial \delta_a} &= \frac{\partial F_{x_b}}{\partial \delta_r} = \frac{\partial F_{y_b}}{\partial \delta_e} = \frac{\partial F_{z_b}}{\partial \delta_a} = \frac{\partial F_{z_b}}{\partial \delta_r} = 0 \\
 \frac{\partial M_{x_b}}{\partial \delta_e} &= \frac{\partial M_{y_b}}{\partial \delta_a} = \frac{\partial M_{y_b}}{\partial \delta_r} = \frac{\partial M_{z_b}}{\partial \delta_e} = 0
 \end{aligned} \tag{7.5.8}$$

The body-fixed components of the gravitational force vector change with aircraft orientation. Since the body-fixed gravitational components only depend on the Euler angles, from Eq. (7.3.15) we have

$$\begin{Bmatrix} W_{x_b} \\ W_{y_b} \\ W_{z_b} \end{Bmatrix} = W \begin{Bmatrix} -\sin(\theta) \\ \sin(\phi)\cos(\theta) \\ \cos(\phi)\cos(\theta) \end{Bmatrix} \tag{7.5.9}$$

which at the equilibrium flight condition gives

$$\begin{Bmatrix} \Delta W_{x_b} \\ \Delta W_{y_b} \\ \Delta W_{z_b} \end{Bmatrix} = \begin{bmatrix} \frac{\partial W_{x_b}}{\partial \phi} & \frac{\partial W_{x_b}}{\partial \theta} & \frac{\partial W_{x_b}}{\partial \psi} \\ \frac{\partial W_{y_b}}{\partial \phi} & \frac{\partial W_{y_b}}{\partial \theta} & \frac{\partial W_{y_b}}{\partial \psi} \\ \frac{\partial W_{z_b}}{\partial \phi} & \frac{\partial W_{z_b}}{\partial \theta} & \frac{\partial W_{z_b}}{\partial \psi} \end{bmatrix} \begin{Bmatrix} \Delta \phi \\ \Delta \theta \\ \Delta \psi \end{Bmatrix} = \begin{bmatrix} 0 & -W \cos \theta_o & 0 \\ W \cos \theta_o & 0 & 0 \\ 0 & -W \sin \theta_o & 0 \end{bmatrix} \begin{Bmatrix} \Delta \phi \\ \Delta \theta \\ \Delta \psi \end{Bmatrix} \tag{7.5.10}$$

By definition, the gravitational force does not produce a moment about the CG.

Using Eqs. (7.5.7) and (7.5.8) in Eqs. (7.5.5) and (7.5.6) and adding Eq. (7.5.10), the small-disturbance force approximations are given by